

Chang-Han Chen

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Education

- **Ph.D. in Theoretical Physics, University of California, Berkeley** expected **end of 2026**
- **B.Sc. in Physics and Mathematics, Massachusetts Institute of Technology** 2022
GPA **4.9/5**; Morse/Orloff award for best senior thesis.

Research Interest

Many challenges in AI, across both capability and safety, are actually about generalization. Think about solving scientific problems beyond human ability, or alignment in unfamiliar situations where reward seems almost impossible.

I aim to improve **data efficiency and generalization** through (1) architectures that use denser computation to reason more effectively, and (2) training algorithms that continue to improve learning from fixed data or rollouts as compute increases. My interests include architectural and optimization changes that enable deeper computation, such as skip connections and preconditioning; diffusion training curricula; ensembling; and stable reinforcement learning for transformers that reason in latent space.

For more on how I think about these questions, see [this blog](#).

Selected Projects

- **MoE autoresearch** 2026
Explored LLM-agent-driven “autoresearch” on autoregressive mixture-of-experts transformers, stacking interventions to a 1.75% bits-per-byte gain that persists under compute-optimal scaling.
- **Optimal block-size curriculum for diffusion language models** 2026 Spring
Designed a block-size curriculum for pretraining block-diffusion language models, sweeping the autoregressive vs. diffusion training budget and finding a consistent sample-efficiency optimum near 30%.
- **A clock is just a way to tell the time** (with G. Penington) 2024 Spring
Applied von Neumann algebra tools to cosmology, clearing up common misconceptions; widely cited in the field.
- **Geometry and entanglement in AdS/CFT and beyond** 2022 Spring
Independent senior thesis, awarded MIT’s Morse/Orloff prize for best senior thesis in theoretical physics.

Honors and Awards

- **Morse/Orloff Research Award** – best senior thesis in theoretical physics, MIT 2022
- **Gold medal, International Physics Olympiad (IPhO)** 2018

References

Prof. Hong Liu	Prof. Geoff Penington	Prof. Aron Wall
MIT	UC Berkeley	U. of Cambridge
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Skills

- **Programming:** Python, PyTorch, JAX, Megatron-LM, SGLang, SkyRL.
- **Research:** von Neumann algebras, random matrix and free probability, language model training.
- **Coursework:** [algorithm design](#), [convex optimization](#), probability theory, statistical learning theory.